. **FORM 2**

THE PATENTS ACT, 1970

(39 of 1970)

AND

The Patents Rules, 2003

COMPLETE SPECIFICATION

(See Section 10 and Rule 13)

1. TITLE OF THE INVENTION:

A SYSTEM AND METHOD FOR A GRAPH-BASED FRAMEWORK FOR REPRESENTING AND ANALYZING PEDIGREE CHARTS

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4. PREAMBLE TO THE DESCRIPTION:

The following specification particularly describes the invention and the manner in which it is to be performed.

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DESCRIPTION OF THE INVENTION:

Technical Field of the Invention

[0001] The present invention relates to genetic and genealogical analysis tools, more specifically, to a structured framework for representing, analyzing, and interpreting pedigree charts, addressing limitations in existing static formats and enhancing the understanding of familial and hereditary relationships.

Background of the Invention

[0002] Pedigree charts have traditionally been employed in genetics, biology, and genealogy to illustrate family connections and inherited characteristics. Nonetheless, conventional representation techniques frequently depend on fixed formats, that are constrained in their capacity to accommodate intricate and changing needs in these areas. These fixed formats limit the thorough depiction of family relationships, impede scalability, and are not adaptable to progress in analytical methods.

[0003] The commonly utilized PED file format, which is structured in tables, has significant limitations, such as its lack of capacity to clearly represent sibling relationships, birth order, and intricate phenotypic characteristics. It also lacks adaptability for iterative modifications or real-time data handling and is unsuitable for relational modeling necessary for dynamic systems. Consequently, researchers encounter inefficiencies, heightened computational complexity, and constraints in precisely modeling hereditary patterns. Hence there is a need for a robust, adaptable, framework capable of dynamically representing familial relationships, incorporating detailed phenotypic and environmental data, and supporting iterative and real-time analyses to overcome the limitations of static pedigree format.

Object of the invention

[0004] A principal objective of the invention is to provide a robust system for accurately representing familial relationships in a flexible, structured framework that overcomes the limitations of traditional static formats such as PED files.

[0005] Another objective of the invention is to enable the explicit representation of sibling relationships, birth order, and detailed phenotypic traits to enhance the analysis of hereditary and genealogical data.

[0006] Another objective of the invention is to facilitate real-time adaptability and iterative updates for dynamic applications, ensuring efficient handling of complex familial structures across diverse fields such as genetics, genealogy, and personalized medicine.

Summary of the Invention

[0007] To the enablement of the present disclosure and related ends, at least one aspect comprises the feature herein after completely described and particularly and/or specifically pointed out in the specification in the section of claims. The following drawings and descriptions outlined in detail enable certain example features of at least one aspect. Described features are indicative, however, of but a few of the many ways in which following principles of various aspects may be employed, and the description is intended to include all such aspects and their equivalents.

[0008] In the present disclosure, the problem of inefficient and static representation of familial relationships and hereditary traits in traditional formats is solved using graph-based system and method for representing and analyzing pedigree charts in a dynamic and flexible manner. The issue of ineffective and unchanging depiction of family connections and inherited characteristics in conventional formats such as the PED file stems from their failure to dynamically reflect intricate relationships and comprehensive phenotypic data. These formats do not offer the flexibility to clearly represent sibling relationships, birth sequence, and multi-trait phenotypes, rendering them inadequate for contemporary analytical requirements.

[0009] Moreover, their fixed nature limits immediate updates, repetitive processing, and adjustment to changing research needs. This restriction obstructs thorough examination in areas such as genetics, ancestry, and tailored healthcare. The invention addresses these issues by presenting a graph-oriented system and technique that enables dynamic and adaptable depiction of pedigree charts. The framework captures intricate relationships, allows real-time updates, and facilitates efficient navigation, thus overcoming the limitations of conventional approaches and improving the usefulness of pedigree data for sophisticated applications.

[0010] The summary provided herein is to introduce a section of concepts in a simple and clear form that is further described in the Detailed Description. The summary provided herein is not intended to particularly identify key features or essential features of the claimed invention or subject matter, nor is it intended to be used as support or as an aid in determining the scope of the claimed subject matter.

[0011] The above summary is descriptive and exemplary only and is not intended to be in any way restricting. In addition to the descriptive aspects, embodiments, and features described in the above summary, further features and embodiments will become apparent by reference to the accompanying drawings and the following detailed description.

Brief Description of Drawings

[0012] The foregoing and other features of embodiments will become more apparent from the following detailed description of embodiments when read in conjunction with the accompanying drawings. In the drawings, like reference numerals refer to like elements.

[0013] In the following description, for the purposes of explanation, numerous specific details are set forth to provide a thorough understanding of the embodiments of the invention. It is apparent, however, to one skilled in the art that the embodiments of the invention may be practiced without these specific details or with an equivalent arrangement. In other instances, well-known structures and devices are shown in block diagram form to avoid unnecessarily obscuring the embodiment of the invention.

[0014] FIG.1 illustrates a network diagram representing a dynamic graph-based pedigree representation system, according to one example embodiment.

[0015] FIG.2 illustrates a system diagram of a dynamic graph-based pedigree representation system, according to one example of the embodiment.

[0016] FIGs.3A-3D illustrates an example scenario representing a dynamic graph-based pedigree representation system, according to one example of the embodiment.

[0017] FIG.4 illustrates a flowchart representing a method for dynamic representation and analysis of pedigree structures, according to one example of the embodiment.

[0018] FIG.5 illustrates a diagram representing the data structure used in the dynamic graph-based pedigree representation system, according to one example of the embodiment.

[0019] FIGs.6A-6D illustrate graphs representing time complexity comparison for family tree representation (V), time complexity comparison for traditional vs. GenetiGraph, time complexity and processing time for nodes and relationships algorithms, and memory access comparison for GenetiGraph and PED format, according to one example of the embodiment.

[0020] TABLE.1 represents a comparative analysis of the PED file format and graph-based pedigree representation system, according to one example of the embodiment.

Detailed Description of the Invention

[0021] Reference will now be made in detail to the description of the present subject matter, one or more examples of that are shown in the figures. Each example is provided to explain the subject matter and not a limitation. Various changes and modifications obvious to one skilled in the art of that invention are deemed within the spirit, scope, and contemplation of the invention.

[0022] As used in this application, the term 'dynamic pedigree representation' refers to various implementations, including (a) frameworks exclusively for structured familial relationship modeling and visualization; (b) combinations of structural frameworks and analytical tools, such as (i) configurations that integrate explicit relationship encoding or trait-based data for enhanced pedigree analysis, or (ii) portions of frameworks combined with modules for phenotype encoding, genealogical analysis, and metadata integration to perform designated functions; and (c) systems that utilize external or embedded analytical processes, even if the algorithms or computational tools are not physically present.

[0023] This definition of 'dynamic pedigree representation' applies throughout this application, including within any claims. For example, the term would encompass implementations involving solely structural modeling, analytical processes, or a combination thereof, such as frameworks with real-time traversal for inheritance pattern visualization or modules for detailed phenotypic data representation. The term further covers systems integrated with interdisciplinary tools in genetics, genealogy, or personalized medicine applications, enabling comprehensive familial analysis and modeling.

[0024] The term "exemplary" is used herein to signify an example, instance, or illustration. Any use of the term 'exemplary' for features such as relational family structures or inheritance pattern encoding should not necessarily be interpreted as preferred or superior to other configurations.

[0025] The term 'content' encompasses genealogical records, phenotypic datasets, and dynamic updates to familial connections. Such content may include resources on sibling relationship encoding, birth order analysis, and metadata integration, enabling researchers and analysts to assess and enhance genealogical insights and hereditary studies.

[0026] As used in this application, the terms 'component,' 'module,' or 'system' refer to combinations of structural frameworks, analytical tools, and related entities forming the foundation of the dynamic pedigree representation framework. For example, a relational modeling solution may include a graph-based framework integrated with tools for phenotype analysis and sibling relationship encoding. These components may operate within a single system or across distributed

configurations, leveraging dynamic traversal capabilities and complex data sets for enhanced familial analysis.

[0027] FIG.1 illustrates a network diagram representing a dynamic graph-based pedigree representation system, according to one example embodiment. The diagram includes the dynamic graph-based pedigree representation system **100**, a remote device **103**, a peripheral component **105,** a local device **107**, and a network **109**. The diagram depicts the structure of the dynamic graph-based pedigree representation system **100**, functioning as an all-encompassing framework for dynamically and flexibly representing and analyzing family relationships. The system overcomes the shortcomings of conventional static formats by utilizing graph-based modeling, permitting real-time updates, and supporting the comprehensive depiction of relationships and phenotypic characteristics.

[0028] At the center lies the dynamic graph-based pedigree representation system **100**, which is tasked with preserving the graph-oriented framework of the pedigree charts. The system offers a flexible and responsive structure that enables the clear representation of sibling connections, birth sequence, and phenotypic information. Utilizing a graph-oriented design, the system guarantees efficient navigation and flexibility for various applications, such as genetics, genealogy, and tailored medicine.

[0029] The components of the system are linked through a network **109** that allows for smooth data transfer among different devices and modules. The network facilitates immediate communication and synchronization, accommodating the system's dynamic characteristics. It serves as the foundation for combining data inputs, processing, and distributing outputs throughout all connected elements.

[0030] The remote device **103** signifies external systems or devices, such as servers or cloud platforms, that either enhance or use the pedigree data for sophisticated analysis and visualization. These remote devices enhance scalability and allow users to access the system from different locations, making the framework ideal for collaborative research and distributed applications.

[0031] The local device **107** is a computing unit or workstation situated close to the central system. It engages directly with the structure for data entry, modification, and local analysis. This element guarantees prompt access for researchers or analysts engaged in pedigree representation activities.

[0032] The peripheral component **105** consists of additional tools or modules that improve the system's functionality. These may comprise tools for integrating metadata, enhancing data, or visual support that offer an improved understanding of family structures and inheritance trends. This element facilitates extra features and guarantees the system’s adaptability to changing research requirements.

[0001] FIG.2 illustrates a system diagram of a dynamic graph-based pedigree representation system, according to one example of the embodiment. The illustration depicts the structure of the dynamic graph-based pedigree representation system, designated as **100**, that includes a processor **201**, a memory **203**, a communication interface **205**, a plurality of sensors **207**, and a power source **209.** The system is structured to carry out dynamic functions by leveraging a mix of processing, storage, communication, sensing, and power features, guaranteeing its flexibility and effectiveness across various applications.

[0033] At the centre of the system lies processor **201**, which acts as the main computational unit. The processor is set up to perform instructions, handle data, and oversee the functions of the different parts of the system. It guarantees the effective management of tasks associated with data analysis, system management, and real-time activities.

[0034] The Memory **203** offers the storage capacity necessary for the system to function. It retains data, instructions, and metadata crucial for the operation of the invention. The memory is set up to store information in an organized manner, allowing for rapid access and effective data retrieval for processing.

[0035] The communication interface **205** enables data transfer between the system and external devices or networks. This element allows for effortless connectivity, facilitating both wired and wireless communication methods to guarantee immediate data transfer and integration with outside systems.

[0036] The system consists of multiple sensors 207 designed to collect data pertinent to the functionality of the invention. These sensors collect environmental or situational data, providing it to the system for assessment and response. Integrating various sensors improves the system's capability to function dynamically and adjust to different situations.

[0037] The energy source **209** supplies the essential power to function the system. It is intended to guarantee continuous operation, supporting multiple power setups, including battery power or external sources, to fulfill the requirements of diverse operational settings.

[0038] FIGs.3A-3C illustrates an example scenario representing a dynamic graph-based pedigree representation system, according to one example of the embodiment. The system includes a node representation unit **301**, a graph-based data processing module **303**, a data storage and retrieval unit **305**, a computational analysis module **307**, and a visualization and reporting unit **309**.

[0039] FIG.3A provides a view of the primary components of the system. It depicts how different modules interact to represent and analyze familial relationships in a graph-based framework.

[0040] The node representation unit **301** forms the foundation of the system, comprising two key elements: the Person Node **301A** and the Relation Node **301B**. The Person Node **301A** represents individuals and stores data such as their identifiers, characteristics, and relationship pointers that link them to other nodes in the system. The Relation Node **301B** serves to establish and store the relationships between Person Nodes **301A**, such as parent-child and sibling relationships. This structure allows the system to efficiently represent complex family trees. Includes a person node **301A** and a relation node **301B**, wherein the person node **301A** stores attributes for each individual and points to familial relationships, and the relation node **301B** stores relational data linking one or more person node **301A.** The relation node **301B** explicitly models sibling relationships by utilizing previous sibling and next sibling pointers, enabling efficient representation of sibling hierarchies and birth order.

[0041] The graph-based data processing module **303** is responsible for processing the relational data stored in the Node Representation Unit **301**. It ensures that the data is updated dynamically, enabling real-time modifications and structural updates within the family tree. This module is integral to maintaining the flexibility and scalability of the system, ensuring that relationships are updated as the family tree grows or changes. Configured to process and dynamically update familial relationship data by traversing the node representation unit **301**, ensuring real-time updates and modifications of data structure. The graph-based data processing module **303** applies near-linear time complexity to ensure efficient processing and retrieval of data as the number of person nodes **301A** increases, thus improving scalability

[0042] The data storage and retrieval unit **305** is designed to store the processed familial data efficiently and allow fast access and retrieval. By optimizing memory usage and utilizing indexed data storage, this unit ensures that relationship data can be accessed quickly for further analysis or modification. Configured to store the processed familial relationship data in a structured graph format, providing for optimized memory utilization and quick retrieval of relationship data through indexed access. The data storage and retrieval unit **305** uses graph-based indexing to facilitate quick data retrieval and minimize memory overhead when processing large datasets, ensuring efficient use of system resources.

[0043] According to some example embodiments of the invention, graph-based indexing refers to the method used to store and retrieve familial relationship data efficiently within the system. This indexing technique organizes Person Node **301A** and Relation Node **301B** in a graph structure, where each node is linked based on familial relationships such as parent-child and sibling connections. The graph-based indexing ensures that relational data is structured in a way that allows for quick traversal and optimized access. By using this indexing technique, the system

reduces the time complexity associated with searching, updating, or adding new relationships in large family trees, thus improving the system's overall efficiency and scalability, particularly when dealing with extensive genealogical datasets.

[0044] The computational analysis module **307** processes the stored data to analyze hereditary relationships, genetic linkages, and other familial patterns. Using graph-based traversal techniques, it efficiently handles complex computations, making it ideal for applications such as genetic prediction or advanced family tree modeling. Configured to process and analyze hereditary relationships, genetic linkages, and relational patterns by applying computational models to extract information from the structured graph format. The computational analysis module **307** integrates one or more graph traversal techniques to improve the accuracy and speed of hereditary pattern detection, thus optimizing the process of genetic analysis.

[0045] According to some example embodiments, graph traversal techniques refer to the methods used to explore and process the nodes and edges within the graph structure. These techniques, such as depth-first search (DFS) and breadth-first search (BFS), allow the system to efficiently navigate through familial relationships stored in Person Nodes (301A) and Relation Nodes (301B). These traversal methods enable real-time updates, the retrieval of related familial data, and the execution of complex analyses, ensuring the system handles large family trees efficiently.

[0046] The visualization and reporting unit **309** provides real-time graphical representations of the family tree or pedigree structure. This unit enhances interpretability by transforming the processed relational data into interactive visual outputs that can be used for diagnostic and analytical purposes, facilitating decision-making and further analysis. Configured to provide a dynamic graph-based pedigree representation of the processed and analyzed hereditary relationships, genetic linkages, and relational patterns thereby enabling users to analyze and interpret the data through interactive visualizations

[0047] FIG.3B includes a node representation unit **301** is broken down into its two essential components: the person node **301A** and the relation node **301B**. the person node **301A** and the relation node **301B**. The person node **301A** signifies individual entities in the pedigree framework, including metadata attributes like gender, name, ID, and phenotypic characteristics. The relation node **301B** represents the connections among two or more person nodes, such as parent-child or sibling links. These nodes constitute the core components of the system, allowing for a dynamic and adaptable depiction of family relationships. The framework enables effective navigation and alteration of the graph structure to meet different research and analytical needs.

[0048] FIG.3C represents a classic illustration of a pedigree chart, representing family connections through generations. The illustration employs conventional symbols, with squares denoting male individuals, circles for females, and lines connecting to show parent-child relationships. The chart emphasizes the structured arrangement of family information, showcasing relationships such as siblings, parents, and children in a visual format. Individuals are arranged from left to right according to birth order in the pedigree analysis, which shows its capacity to incorporate extra metadata, such as birth sequence, phenotypic characteristics, or environmental influences.

[0049] FIG.3D includes a graph-oriented illustration of the pedigree arrangement, highlighting the interactive and relational structure presented by the innovation. In this illustration, every person node (e.g., GF1, GM1, Fthr, Mthr) is linked to related relation nodes (e.g., FRel, MRel) via directional and bidirectional edges, indicating family ties like parent-child and sibling relations. The graph framework allows for the representation of intricate relationships, like the order of sibling births (e.g., C1, C2, C3), and facilitates the incorporation of phenotypic and environmental metadata. The directed edges enhance navigation and evaluation, guaranteeing computational efficiency and flexibility for dynamic updates and iterative modeling. The representation overcomes the shortcomings of conventional charts by offering a strong, adaptable, and expandable framework for analyzing pedigree data.

[0050] FIG.4 illustrates a flowchart representing a method for dynamic representation and analysis of pedigree structures, according to one example of the embodiment. Demonstrates a sequential procedure for executing the dynamic graph-based pedigree representation system **100**, outlining the functionality and workings of its main components.

[0051] At step **401**, the method includes creating a person node **301A** with basic details such as identifiers, attributes, and relationship pointers followed by establishing a relation node **301B** to define how different family members are connected within a node representation unit **301**.A Person Node **301A** is configured within the node representation unit **301**. the person node **301A** stores various identifiers, attributes, and relationship pointers that represent an individual within the family structure. This data is essential for defining familial relationships, such as parent-child and sibling relationships. Following this, a relation node **301B** is established, which connects multiple person nodes **301A**, thereby representing familial relationships between individuals, such as parents and children or siblings. This step creates the foundational structure of the family tree and enables the system to dynamically represent family relationships.

[0052] At step **403**, the method includes processing and dynamically updating relational data obtained from the node representation unit **301** via a graph-based data processing module **303** to enable real-time modifications and structural updates. Once the person nodes **301A** and relation nodes **301B** are established, the graph-based data processing module **303** processes and dynamically updates the relational data obtained from the node representation unit **301**. this processing is done in real-time, ensuring that as new relationships or modifications are made to the family tree, the system automatically updates the data structure. the graph-based data processing module **303** allows for structural updates, ensuring that the family tree remains accurate and up to date as data is added or modified.

[0053] At step **405**, the method includes encoding **405**, sibling relationships and birth order by assigning structured traversal pointers within the person node **301A**, based on processed relational data, to understand and track family connections accurately. This step involves encoding sibling relationships and birth order within the Person Node **301A**. By assigning structured traversal pointers within the Person Node **301A**, the system efficiently navigates between siblings in a family. These pointers help represent the order of birth and sibling connections within the family tree. The encoding of these relationships is based on the processed relational data from the previous step, ensuring accurate and efficient traversal of the family tree structure, as well as the correct representation of familial dynamics.

[0054] At step **407**, the method includes storing and retrieving encoded sibling relationships and birth order data via a data storage and retrieval unit **305** for optimized memory utilization and structured data access. The encoded sibling relationships and birth order data are stored and retrieved through the Data Storage and Retrieval Unit **305**. This unit ensures that the encoded data is stored in an efficient manner, with optimized memory utilization. The structured storage and retrieval processes enable quick access to familial relationship data as needed, ensuring that the system can manage and access large datasets without compromising performance. This also enables smooth interaction with the rest of the system, as it can dynamically pull relational data when required.

[0055] At step **409**, the method includes executing a computational analysis using a computational analysis module **307** to examine hereditary relationships, genetic connections, and predictive trends using advanced graph-based analysis techniques. The Computational Analysis Module **307** is used to process the stored data, analyzing hereditary relationships, and genetic linkages, and perform predictive modeling. This step leverages the graph structure created by Person Node **301A** and Relation Node **301B** to apply computational techniques that analyze genetic patterns and family dynamics. The system uses graph-based traversal techniques to ensure efficient processing, allowing for the analysis of complex familial data, such

as genetic inheritance and other hereditary patterns, based on the relationships encoded within the family tree.

[0056] At step **411**, the method includes generating real-time graph-based visualizations of a pedigree structure for analytical and diagnostic purposes, wherein the visualization and reporting unit **309** processes executed data to enhance interpretability and decision-making. The Visualization and Reporting Unit **309**, generates real-time graph-based visualizations of the family tree or pedigree structure. This allows users to visually explore the familial relationships and the results of the computational analysis. The visualizations enhance interpretability and decision-making, especially for analytical or diagnostic purposes. This step is crucial for users to quickly understand complex familial data and make informed decisions based on the system's analysis. The Visualization and Reporting Unit **309** processes the data generated from the previous steps to produce graphical outputs that are easy to interpret and use.

[0057] According to some example embodiments, dynamically updating familial relationships within the graph structure ensures that the family tree remains accurate and up-to-date as new information or modifications are introduced. This process involves continuously tracking and incorporating changes, such as the addition of new family members, updated relationship status, or changes in existing connections (e.g., sibling order or parent-child links). By utilizing graph-based traversal techniques, the system can efficiently update the connections between person nodes **301A** and relation nodes **301B** in real-time. This capability ensures that any new familial data is seamlessly integrated, maintaining the integrity and consistency of the family tree without requiring manual intervention, which is particularly crucial for applications in genealogical research and genetic analysis.

[0058] FIG.5 illustrates a diagram representing the data structure used in the dynamic graph-based pedigree representation system, according to one example of the embodiment.

[0059] Central to the system are the relation of origin components of the Father and Mother. These signify the relationship between every individual node and their guardians. The Father and Mother structures both include a pointer to the origin relation, which points to their parental figures, along with a pointer to the first child. This structured arrangement guarantees that family ties and ancestry are effectively showcased, with the framework illustrating the direct connection of each parent to their children. Additionally, the parent nodes contain references to the preceding and following siblings, facilitating a quick navigation of sibling connections, which is crucial for grasping the dynamics among various children in a family.

[0060] Alongside direct parent-child connections, the framework also encompasses the relationships among siblings. For instance, the Father's Elder Sibling and the

Mother's Younger Sibling are individually represented in the framework. These sibling nodes also hold relation pointers, and the sibling pointers (i.e., previous sibling and next sibling) in these nodes facilitate a structured and efficient depiction of the sibling hierarchy. This guarantees that not only the direct links between parents and children are recorded, but also the broader connections among relatives like aunts, uncles, and cousins.

[0061] The system also directly manages the children in the family. The First Child, Middle Child, and Last Child nodes signify each child's distinct role within the family. These nodes hold identical relationship pointers, connecting each child to their source relation and offering previous sibling and next sibling pointers for streamlined navigation. This framework not only facilitates the depiction of birth sequence but also permits the system to handle the whole sibling cohort in a sequential manner. By maintaining this relational structure, the system ensures accurate and efficient navigation through the family tree, making it ideal for genealogical studies, medical research, and AI-based familial analysis.

[0062] FIGs.6A-6D illustrate graphs representing time complexity comparison for family tree representation (V), time complexity comparison for traditional vs. GenetiGraph, time complexity and processing time for nodes and relationships algorithms, and memory access comparison for GenetiGraph and PED format, according to one example of the embodiment.

[0063] FIG. 6A illustrates the comparison between GenetiGraph (O(V)) and the traditional PED Format (O(V + E)) based on the number of people in the family tree (V). As shown, GenetiGraph grows linearly (O(V)), meaning the time taken increases proportionally with the number of individuals. This reflects the efficient processing of each individual node. On the other hand, the PED Format grows at a faster rate due to the additional complexity of handling edges (E), representing the relationships in the family tree. The PED Format becomes significantly slower as the tree grows, especially as the number of relationships increases. The gap between the two lines highlights GenetiGraph's superiority in terms of scalability and performance efficiency for large datasets.

[0064] FIG. 6B, the time complexity of GenetiGraph (O(V)) and the PED Format (O(V + E)) are compared against the number of individuals (V) in the family tree. The GenetiGraph algorithm continues to scale linearly, indicating that the time taken increases steadily as the number of individuals increases. However, PED Format shows a much steeper increase in processing time due to the additional edge processing. This is because the PED Format requires handling relationships for every individual, which results in a higher computational cost. As the number of relationships grows, PED Format becomes increasingly inefficient, further emphasizing GenetiGraph's advantage in dealing with large genealogical datasets.

[0065] FIG. 6C compares the time taken for both GenetiGraph and PED Format as the number of individuals (V) increases, while also accounting for the relationships (E) in the family tree. As expected, GenetiGraph continues to exhibit linear time complexity (O(V)), meaning its processing time increases only with the number of nodes in the tree, maintaining consistent efficiency. In contrast, PED Format experiences an exponential increase in time due to the added complexity of handling the edges (E), which increases with the number of relationships. The graph clearly shows that the PED Format becomes much more computationally expensive as the family tree grows larger, making it less suitable for large-scale genealogical research.

[0066] FIG. 6D compares the processing time of GenetiGraph (O(V)) and PED Format (O(V + E)) when processing nodes (V) in the family tree. GenetiGraph continues to scale efficiently with linear processing time (O(V)), indicating that it handles large datasets with consistent performance. On the other hand, PED Format experiences a significant increase in processing time due to the additional overhead involved in scanning both nodes and edges. This exponential growth in time reflects the inefficiency of the PED Format, particularly when managing large family trees with extensive relationship data. GenetiGraph's ability to directly access and process data in linear time (O(V)) makes it more suitable for applications requiring fast processing of large datasets, such as AI-based analyses and real-time genealogical studies. Here from **FIG.6A to FIG.6B** GenetiGraph refers to the dynamic graph-based pedigree representation system **100**.

[0067] TABLE.1 represents a comparative analysis of the PED file format and graph-based pedigree representation system, according to one example of the embodiment.

[0068] The following table presents a comparative analysis between the Existing PED File Format and the New Graph-Based Pedigree Representation System, highlighting key technical advancements. The comparison is strictly based on data from the Information Disclosure Form (IDF), ensuring accuracy and relevance to the invention's improvements.

[0069] In traditional PED file formats, memory traversal is not possible, limiting data access and computational efficiency. Additionally, only a single characteristic can be encoded per entity, making it incompatible with multi-phenotypic analysis. The PED format also requires extensive processing overhead, as all data must be scanned to access specific relationships. Furthermore, it operates as a static format, preventing real-time updates or data augmentation. The lack of explicit relationship modeling further increases error rates in complex genealogical studies.

[0070] The Graph-Based Pedigree Representation System, as disclosed in the invention, introduces efficient memory traversal and supports encoding multiple

characteristics with a 5x-10x improvement over existing formats. It significantly reduces memory access time (by 50-70%) through structured indexing and optimized traversal mechanisms. The new format also allows dynamic updates, with up to 60% faster data augmentation, making it adaptable to real-time genealogical and medical applications.

[0071] One of the most notable advantages is its seamless integration with AI-based models, eliminating the need for separate preprocessing and reducing AI-related computational overhead by 80-90%. Furthermore, the graph-based structure is highly scalable, efficiently handling datasets exceeding 10 million records. The traversal process operates in near-linear time, achieving a 70-90% reduction in data access requirements, significantly improving computational speed and efficiency. Moreover, the system introduces explicit relationship modeling, reducing errors by 30-50%, particularly in sibling and birth order relationships.

[0072] According to some example embodiments, a significant improvement in the representation, analysis, and processing of familial relationships within genealogical data. By employing a graph-based data structure, the invention efficiently models complex family trees, capturing not only direct parent-child relationships but also intricate sibling and extended family connections. The dynamic traversal of the graph enables real-time updates and modifications to familial data, providing faster and more accurate processing compared to traditional formats. The use of relation nodes and sibling pointers allows for optimized memory utilization and quick access to genealogical relationships, significantly reducing time complexity and memory overhead. Furthermore, the system's compatibility with AI and machine learning algorithms enhances its ability to perform predictive analysis and genetic modeling, offering substantial improvements in scalability, data handling, and computational efficiency over conventional methods. This results in a highly efficient, adaptable system that can scale to handle large datasets while ensuring precise and consistent representation of complex family structures.

[0073] According to some example embodiments, the innovation offers multiple technical and business benefits, improving its usefulness and relevance in different fields. From a technical standpoint, the dynamic graph-based pedigree representation system **100** shows adaptability by accommodating changing research needs, such as incorporating new metadata or attributes, and maintaining its significance in fluid and cross-disciplinary areas. Its cross-disciplinary relevance renders it adaptable for applications in genetics, computational biology, personalized medicine, and genealogy. Moreover, the framework is easy to use, making data exploration and visualization simpler, greatly enhancing usability for researchers. The system additionally facilitates data augmentation, allowing for the creation of enhanced datasets for sophisticated machine learning and predictive

modeling. From a business standpoint, the innovation presents market differentiation by delivering a better option compared to obsolete formats such as PED, attracting interest from both researchers and commercial organizations. Broadened applications encompass uses in genealogical services, medical diagnostics, and AI-enhanced genetic studies. In addition, the innovation is economical, lowering computing and storage expenses in extensive applications, and is post-proof, featuring an adaptable framework that guarantees prolonged usefulness and flexibility in both academic and business settings.

[0074] According to some example embodiments, the innovation brings notable improvements in the representation and examination of pedigree data, targeting enduring limitations found in current formats such as the PED file. A significant technical benefit is found in its clear encoding of relationships, enabling direct representation of sibling connections and birth sequence, which must usually be deduced. The system additionally offers thorough phenotypic representation, allowing for the encoding of various traits and environmental factors for each individual, thus surpassing the single-phenotype limitation of conventional formats. The framework enables dynamic updates, permitting real-time data addition and modification, thus enhancing iterative analysis and flexible modeling. Moreover, the invention utilizes an improved graph-based framework that boosts computational efficiency, minimizes redundancy, and allows for quicker traversal than flat tabular structures. Tailored for AI/ML compatibility, the system works smoothly with graph-based techniques such as graph neural networks, improving predictive accuracy and simulation features. Its scalability also guarantees efficient management of extensive, intricate datasets while incurring minimal computational costs.

[0075] According to some example embodiments, the graph-oriented data structure significantly differs from conventional formats, allowing for a flexible and scalable design that facilitates the simple addition or removal of nodes and metadata. This adaptability enables the system to meet changing research requirements. The innovation is tailored for AI/ML integration, utilizing graph-based techniques for sophisticated analysis. The innovative aspect is its capacity to dynamically encode sibling relationships and birth order, a capability that has no equivalent in current formats. The invention enhances computational abilities markedly by converting a static, flat structure into a dynamic, interconnected system. Moreover, the system's interdisciplinary approach merging genetics, computational biology, and machine learning knowledge distinctly sets it apart as a unique and creative solution in the realm of pedigree representation and analysis.

[0076] According to some example embodiments, the Dynamic Graph-Based Pedigree Representation System allows for efficient and scalable construction of family trees. It dynamically updates and represents familial relationships, enabling

users to explore direct and extended family connections across multiple generations. This system enhances accuracy and depth in ancestry research, as it can incorporate new data seamlessly, reflecting updates such as newly discovered relatives or changes in family structure.

[0077] According to some example embodiments, the Dynamic Graph-Based Pedigree Representation System aids in tracking and analyzing hereditary health conditions. By modeling genetic relationships within the family tree, the system provides insights into the inheritance patterns of genetic disorders. Healthcare professionals can use this information to predict potential genetic risks and personalize treatment plans. The system's real-time updates ensure that any new medical information or genetic data is immediately reflected in the family tree, enhancing diagnostic accuracy.

[0078] According to some example embodiments, the Dynamic Graph-Based Pedigree Representation System enables the integration of artificial intelligence (AI) to study genetic linkages and predict hereditary traits. The system can process large datasets of genetic information, identifying patterns across family trees. By continuously updating the graph-based model as new data is gathered, it helps researchers to efficiently analyze genetic relationships, identify new genetic markers, and further understand complex genetic traits. The system's ability to scale and handle evolving data makes it a crucial tool in advancing genetic research.

[0079] The many features and advantages of the invention are apparent from the detailed specification, and thus, it is intended by the appended claims to cover all such features and advantages of the invention that fall within the true spirit and scope of the invention. Further, since numerous modifications and variations will readily occur to those skilled in the art, it is not desired to limit the invention to the exact construction and operation illustrated and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the invention.

We claim,

1. A dynamic graph-based pedigree representation system **100,** comprising:
   a) a node representation unit **301**, includes a person node **301A** and a relation node **301B**, wherein the person node **301A** stores attributes for each individual and points to familial relationships, and the relation node **301B** stores relational data linking one or more person node **301A**;
   b) a graph-based data processing module **303**, configured to process and dynamically update familial relationship data by traversing the node representation unit **301**, ensuring real-time updates and modifications of data structure;
   c) a data storage and retrieval unit **305**, configured to store the processed familial relationship data in a structured graph format, providing for optimized memory utilization and quick retrieval of relationship data through indexed access;
   d) a computational analysis module **307**, configured to process and analyze hereditary relationships, genetic linkages, and relational patterns by applying computational models to extract information from the structured graph format; and
   e) a visualization and reporting unit **309**, configured to provide a dynamic graph-based pedigree representation of the processed and analyzed hereditary relationships, genetic linkages, and relational patterns thereby enabling users to analyze and interpret the data through interactive visualizations.

   wherein the node representation unit **301** contains the person node **301A** and the relation node **301B** to organize family relationships, the graph-based data processing module **303** updates and processes the family relationship data in real time, the data storage and retrieval unit **305** efficiently stores and retrieves relationship data, the computational analysis module **307** examines genetic links and patterns, while the visualization and reporting unit **309** creates clear, real-time graphs for better understanding and decision-making.
2. The system **100** as claimed in claim 1, wherein the graph-based data processing module **303** dynamically links one or more person nodes **301A** and one or more relation nodes **301B** using directional and bidirectional connections to represent familial relationships.
3. The system **100** as claimed in claim 1, wherein the data storage and retrieval unit **305** uses graph-based indexing to facilitate quick data retrieval and minimize memory overhead when processing large datasets, ensuring efficient use of system resources.
4. The system **100** as claimed in claim 1, wherein the computational analysis module **307** integrates one or more graph traversal techniques to improve

the accuracy and speed of hereditary pattern detection, thus optimizing the process of genetic analysis.

5. The system **100** as claimed in claim 1, wherein the relation node **301B** explicitly models sibling relationships by utilizing previous sibling and next sibling pointers, enabling efficient representation of sibling hierarchies and birth order.
6. A method for dynamic graph-based pedigree representation, comprising:
   a) creating **401**, a person node **301A** with basic details such as identifiers, attributes, and relationship pointers followed by establishing a relation node **301B** to define how different family members are connected within a node representation unit **301**;
   b) processing **403**, and dynamically updating relational data obtained from the node representation unit **301** via a graph-based data processing module **303** to enable real-time modifications and structural updates;
   c) encoding **405**, sibling relationships and birth order by assigning structured traversal pointers within the person node **301A**, based on processed relational data, to understand and track family connections accurately;
   d) storing and retrieving **407**, encoded sibling relationships and birth order data via a data storage and retrieval unit **305** for optimized memory utilization and structured data access;
   e) executing **409**, a computational analysis using a computational analysis module **307** to examine hereditary relationships, genetic connections, and predictive trends using advanced graph-based analysis techniques; and
   f) generating **411**, real-time graph-based visualizations of a pedigree structure for analytical and diagnostic purposes, wherein the visualization and reporting unit **309** processes executed data to enhance interpretability and decision-making.
7. The method as claimed in claim 6, wherein the graph-based data processing module **303** applies near-linear time complexity to ensure efficient processing and retrieval of data as the number of person nodes 301A increases, thus improving scalability.
8. The method as claimed in claim 6, wherein the visualization and reporting unit **309** generates real-time graphical representations of the family tree, enabling interactive exploration and analysis of the familial structure and its genetic implications.
9. The method as claimed in claim 6, further comprises a step of dynamically updating familial relationships within the graph structure to reflect new information or modifications, ensuring that the family tree remains accurate and up-to-date.

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Dated this on 24th day of March 2025

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ABSTRACT

A SYSTEM AND METHOD FOR A GRAPH-BASED FRAMEWORK FOR REPRESENTING AND ANALYZING PEDIGREE CHARTS

The invention presents an innovative graph-based system for the dynamic depiction and evaluation of pedigree charts, overcoming the constraints of conventional static formats such as the PED file. The framework clearly defines family relationships, such as sibling ties and birth sequence, and enables thorough phenotypic depiction by including various traits and environmental influences for each individual. Its dynamic structure facilitates real-time changes and updates, allowing for iterative analysis and flexible modeling. Utilizing a graph-oriented framework, the system improves computational effectiveness, minimizes redundancy, and speeds up navigation. The framework is designed for seamless integration with various techniques, including graph neural networks, to enhance predictive accuracy and simulation efficiency. Furthermore, the invention is very scalable, able to manage extensive and intricate datasets with little computational burden. This interdisciplinary approach integrates genetics, computational biology, and machine learning, resulting in a flexible resource for use in research, healthcare, genealogy, and personalized medicine.

FIG.3A

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